

Assignment 2

Program : Choose an item.

Date : 20 April 2026

Examiners : Dr. Mohamed Elsayed

Course/Code : Statistics for Science/PHM311

Example 1: For two events A and B , let $P(A|B) = 0.4, P(B|A) = 0.25$

and $P(B \cap A) = 0.12$.

- (i) Calculate the value of $P(B)$.
- (ii) Give a reason why A and B are not independent.
- (iii) Calculate the value of $P(B^c \cap A)$.

Example 2: Three newspapers A, B, C are published in a city and a survey of readers indicates the following: 20% read A, 16% read B, 14% read C, 8% read both A and B, 5% read both A and C, 4% read both B and C, and 2% read all three. For a person chosen at random, find the probability that he reads none of the papers.

Example 3: Fifty-two percent of the students at a certain college are females. Five percent of the students in this college are majoring in computer science. Two percent of the students are females majoring in computer science. If a student is selected at random, find the probability that:

- (i) This student is female, given that the student is majoring in computer science.
- (ii) (ii) This student is majoring in computer science, given that the student is female.

[illegible]

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Example 4: Two video products distributors supply video tape boxes to a video production company. Company A sold 100 boxes of which 5 were defective. Company B sold 300 boxes of which 21 were defective. If a box was defective, find the probability that it came from Company B.

Example 5: Box I contains 6 green marbles and 4 yellow marbles. Box II contains 5 yellow marbles and 5 green marbles. A box is selected at random and a marble is selected from the box. If the marble is green, find the probability it came from Box I.

[illegible]

